

C. Triple Removal

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

Tired of supporting ranged carries, Keria is now creating a data structure problem about supporting range queries.

For an array $b = [b_1, b_2, \dots, b_m]$ of length m where $b_i = 0$ or $b_i = 1$, consider the following **triple removal** operation:

1. Choose three indices $1 \leq i < j < k \leq m$ such that the elements at these positions are identical ($b_i = b_j = b_k$).
2. Remove these three elements from the array. The cost of this operation is defined as $\min(k - j, j - i)$. After the removal, the remaining parts of the array are concatenated, and their indices are relabeled.

We want to make the array b empty using the **triple removal** operation. Hence, we define the *total cost* of an array as the minimum possible sum of the costs of **triple removal** operations required to make the array empty. If it is impossible to make the array empty, the cost is defined to be -1 .

Keria wants to test his data structure. For this, you must answer q independent queries. Initially, you are given an array $a = [a_1, a_2, \dots, a_n]$ of length n where $a_i = 0$ or $a_i = 1$. For each query, you are given a range $1 \leq l \leq r \leq n$ and must find the cost for the array $[a_l, a_{l+1}, \dots, a_r]$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \leq n, q \leq 250\,000$) — the length of the array and the number of queries.

The next line contains n integers $a_1, a_2, \dots, a_n$ ($a_i = 0$ or $a_i = 1$) — the elements of the array.

Then q lines follow. The i -th of them contains two integers l_i and r_i ($1 \leq l_i \leq r_i \leq n$) — the range of the subarray for the i -th query.

It is guaranteed that the sum of n over all test cases does not exceed 250 000.

It is guaranteed that the sum of q over all test cases does not exceed 250 000.

Output

For each test case, output q lines. The i -th line should contain a single integer representing the answer to the i -th query.

Example

input

Copy

```
2
12 4
0 0 1 1 0 1 0 1 0 1 1 0
1 12
2 7
5 10
6 11
6 3
0 0 0 1 1 1
1 3
4 6
```

Squarepoint Challenge (Codeforces Round 1055, Div. 1 + Div. 2)

Contest is running

01:29:48

Contestant



→ Submit?

 Language: Python 3.13.2 
 Almost always, if you send a solution on PyPy, it works much faster

 Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	380
Problem B	1140
Problem C	1140
Problem D	1900
Problem E	2280
Problem F	3040
Problem G	3040
Problem H1	3040
Problem H2	1900
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 01:30 from the first attempt

1 6

output

Copy

4
2
3
-1
1
1
2

Note

Explanation of the first test case, first query (1 12):

The subarray is $[0, 0, 1, 1, 0, 1, 0, 1, 0, 1, 1, 0]$. There are six 0s and six 1s. A possible optimal sequence of operations is:

1. Remove the three 1s at indices 3, 4, 6. The cost is $\min(6 - 4, 4 - 3) = \min(2, 1) = 1$.
The array becomes $[0, 0, 0, 0, 1, 0, 1, 1, 0]$.
2. Remove the three 0s at indices 1, 2, 3. The cost is $\min(3 - 2, 2 - 1) = \min(1, 1) = 1$.
The array becomes $[0, 1, 0, 1, 1, 0]$.
3. Remove the three 1s at indices 2, 4, 5. The cost is $\min(5 - 4, 4 - 2) = \min(1, 2) = 1$.
The array becomes $[0, 0, 0]$.
4. Remove the three 0s at indices 1, 2, 3. The cost is $\min(3 - 2, 2 - 1) = \min(1, 1) = 1$.
The array is now empty.

The total cost is $1 + 1 + 1 + 1 = 4$.

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